

# Multiplying a polynomial by a monomial



1

a  $r^2 + 5r$

c  $y^2 - 9y$

e  $2t^2 - 14t$

g  $2n^2 - 28n$

i  $-15t^2 - 40t$

k  $-2m^2 - 48m$

m  $7y^3 + 56y$

o  $70u^6 + 80u^5$

q  $7wy^2 + 7w^2y$

s  $12a^2 + 14ab - 8a$

u  $-4a^3b^3 - 16a^2b^2 + 24ab$

b  $w + w^2$

d  $-m^2 - m$

f  $m - m^2$

h  $30y^2 - 60y$

j  $4.37y^2 + 9.2y$

l  $y^2 + 5y$

n  $-15y^3 + 10y^2$

p  $54u^{14} + 54u^{13}$

r  $8a^2 - 32ab - 40ac$

t  $10a^3 + 4a^2 + 6a$

v  $-10a^4b^2 - 25a^3b^2 - 10ab^2$

2

a  $y^2 - 4y + 10$

c  $80u^2 + 8uv - 5$

e  $x^2 + 26x + 12$

g  $u^2 - 7u - 72$

i  $7y^2 - 31y$

k  $-2y^2 + 23y$

m  $34y^2 + 44y$

o  $29p^2 - 10pq$

q  $9wy^2 + 5w^2y + 8wy$

s  $-5y^3 + 17y^2 + 3$

u  $n^3 - 6n^2 - 16n$

w  $15z^3 - 4z^2 + 24z$

y  $6x^2 - 12x - 2$

b  $38y^2$

d  $x^2 + 10x + 30$

f  $9y^2 + 62y - 2$

h  $4y^2 + 48y + 5$

j  $-48u^2 + 24uv$

l  $25x^2 + 28x$

n  $11a^2 + 8ab$

p  $7u^2 + 10uv$

r  $44x^2 + 64xy$

t  $-y^3 + 38y$

v  $20x^3 - 52x^2$

x  $75b^2$

z  $10c^4 - 10c^3 + 11c^2 + 4c - 6$

3  $45u^2v + 45uv^2$

4  $(50p^2q - 35pq^2) \text{ cm}^2$

5  $\frac{1}{5}x^5 - \frac{1}{5}x^5$



## Additional questions

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a  $x^2 - 5x + 6$

b  $ab - 6a + 9b - 54$

c  $42y^2 + 6y - 36$

d  $9y^2 + 90y + 144$

e  $5v^3 + 22v^2 - 20v - 25$

f  $m^3 - m^2 - 23m + 15$

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a If  $x$  feet represents the width of the garden, then the area will be  $x(2x - 1) = 2x^2 - x$  square feet.

b  $(x + 4)(2x + 3) - x(2x - 1) = 10x + 12$

c A garden with a width of 7' and a length of 13' will need 96 square feet of pavers to create a 2' border.